

Updating the Power Sector Competitiveness Dashboard Simulator

When you need to add or remove indicators from your dashboard, follow these steps in `indicator_formatting.R`:

Step 1: Update the Base Data CSV

Edit `state_indicator_data.csv`:

- **Add columns** for new indicators with state-by-state values (0-1 scale)
- **Remove columns** for deprecated indicators

Step 2: Update `indicator_steps` (line 17)

Define the slider step size for each indicator:

```
indicator_steps <- c(
  "Your New Indicator Name (C)" = 0.25, # Add new indicators
  "Third Party PPA (C)" = 0.5,
  # ... existing indicators ...
  # Remove any deprecated indicators
)
```

Rules:

- Use (C) for Consumer, (S) for Structure, (M) for Regional Market
- Common step values: 0.25, 0.33, 0.5, 1.0
- Binary indicators (yes/no) → use 1.0

Step 3: Update `indicator_groups_lookup` (line 38)

Adjust the counts in the `rep()` function to match your new totals:

```
indicator_groups_lookup <- data.frame(
  Indicator = names(indicator_steps),
  Group = c(
    rep("Consumer (C)", 5),    # ← Update count if you added/removed Consumer indicators
    rep("Structure (S)", 7),    # ← Update count if you added/removed Structure indicators
  )
)
```

```

    rep("Regional Market (M)", 3) # ← Update count if you added/removed Market indicators
  ),
  stringsAsFactors = FALSE
)

```

Important: The sum of these three numbers must equal length(indicator_steps).

Step 4: Update indicator_info table (line 49)

Add short descriptions for new indicators or remove old ones:

```

indicator_info <- list(
  "Your New Indicator Name (C)" =
    "Brief description of what this indicator measures and why it matters.",

  # ... existing indicators ...
)

```

Step 5: Update indicator_scoring_details table (line 101)

Add scoring criteria for new indicators or remove old ones:

```

indicator_scoring_details <- list(
  "Your New Indicator Name (C)" =
    "1.0: Full description.<br>0.5: Partial description.<br>0.0: None.",

  # ... existing indicators ...
)

```

Note: Use
 for line breaks in the HTML table display.

Step 6: Run the Script

Execute indicator_formatting.R to regenerate the revised RDS files:

- state_indicator_data.rds
- indicator_data.rds

Step 7: Verify

The Shiny app will now **automatically**:

- Create sliders for all indicators
- Update category counts in labels ("Consumer (C): (X indicators)")
- Update the weight distribution display
- Build detail tables with new descriptions
- Calculate scores correctly

Quick Checklist

When adding an indicator:

- Add column to state_indicator_data.csv
- Add entry to indicator_steps
- Update count in indicator_groups_lookup
- Add entry to indicator_info
- Add entry to indicator_scoring_details
- Run the indicator_formatting.R script to generate RDS files
- Test in Shiny app

When removing an indicator:

Remove column from state_indicator_data.csv

- Remove entry from indicator_steps
- Update count in indicator_groups_lookup
- Remove entry from indicator_info
- Remove entry from indicator_scoring_details
- Run script to generate RDS files
- Test in Shiny app

No changes needed in the main Shiny app code (scenario_simulator_PSCD.r)

All counts update dynamically via indicator_counts.